

Smil Labs

Data Storm 2026 - Latent Outlet Potential Estimation Final Technical Report

Item	Value
Target month	January 2026
Platform output	Results/smil_labs_predictions.csv
Full business output	Results/smil_labs_predictions_full_20000.csv
Primary notebook	Notebooks/01_latent_potential_pipeline.ipynb
POI notebook	Notebooks/03_poi_enrichment.ipynb

Objective

Estimate the latent maximum monthly purchase potential in liters for traditional trade outlets. Historical sales are treated as censored observations, because observed volume is the minimum of true consumer demand and operational constraints such as delivery caps, stockouts, cooler availability, or credit limits.

Core equation

$$\text{Observed Volume} = \min(\text{True Consumer Demand}, \text{Operational Constraint Ceiling})$$

Our approach uses observed maximum volume as a defensible lower bound, then estimates uncapped upside from peer frontiers, outlet structure, spatial catchment intensity, and external POI footfall signals.

1. Lakehouse Pipeline and Data Forensics

The solution follows a Bronze, Silver, and Gold local lakehouse architecture. Raw files are preserved in Bronze; reusable quality checks and dataset-specific cleaning produce Silver; Gold contains modeling features, predictions, POI features, and diagnostics.

Layer	Purpose	Key outputs
Bronze	Raw ingestion, no transformations	raw CSV copies, ingestion audit
Silver	Validated and cleaned data	clean CSVs, rejected record stores
Gold	Feature/model layer	outlet features, POI features, diagnostics, predictions

Reusable data quality checks

- Duplicate checks on configurable keys.
- Mandatory null and blank checks.
- Referential integrity checks between outlets, coordinates, and transactions.
- Numeric range checks for volume, bill value, coordinates, year, month, and cooler count.
- Categorical domain checks for outlet type, outlet size, distributor, and seasonality labels.

Main trapped anomalies

Finding	Count / impact
Missing outlet size	196 rows standardized to Unknown
Lowercase outlet size small	600 rows standardized to Small
Outlet type Grocry	390 rows standardized to Grocery
Outlet type Bakry	395 rows standardized to Bakery
Invalid coordinate rows	240 coordinate rows, producing 480 rejected check records
Rejected transaction rows	9,606 rejected check records from non-positive volume/bill checks

Rejected records are written to data/silver_rejected/ with dataset name, failed check, and failure reason. This prevents silent data loss while keeping the model-ready layer clean.

2. EDA, Catchment Features, and POI Acquisition

Full-dataset EDA used 20,000 outlet master records and 2,376,389 transaction rows across 2023-2025. All 20,000 outlets appear in transaction history. The median outlet maximum observed monthly volume is 164.0 L, while the 95th percentile is 1,307.9 L, showing a large performance frontier gap.

EDA metric	Value
Outlet records	20,000
Transaction rows	2,376,389
Distinct months	36
Distributors	10
SKUs	10
Median observed max monthly liters	164.0 L
95th percentile observed max monthly liters	1,307.9 L

Internal catchment features

Using valid outlet coordinates, the pipeline creates outlet density features with a haversine BallTree: outlet counts within 1 km, 2 km, and 5 km; same-type count within 2 km; same-distributor count within 5 km; nearest outlet distance; and a catchment density score.

External POI acquisition

POI enrichment uses OpenStreetMap through the Overpass API. To avoid a single oversized request, queries are split by category and cached under data/bronze/. Parsed POIs are stored in data/gold/poi_cleaned.csv and outlet-level POI features in data/gold/outlet_poi_features.csv.

POI category	Examples
Education	schools, colleges, universities, kindergartens
Transport	bus stops, stations, railway stops
Market / retail	supermarkets, convenience shops, marketplaces
Healthcare	hospitals, clinics, pharmacies
Food service	restaurants, cafes, fast food, bakeries
Office / finance	banks, ATMs, offices
Religious and tourism	places of worship, hotels, attractions

Latest POI run: 902 valid-coordinate target outlets, 9,581 parsed OSM POIs, mean 15.24 POIs within 1 km and 59.82 within 2 km.

3. Latent Potential Methodology

The model is designed as a guarded uncap framework rather than a standard sales forecast. Observed history provides a demonstrated floor; comparable high performers and demand signals define the ceiling.

Lower bound

```
lower_bound = max(historical_max, january_max, recent_3_month_max)
```

The final prediction is never allowed below this bound.

Frontier and constraint logic

- A gradient boosting model estimates observed monthly behavior.
- A 90th percentile quantile gradient boosting model estimates a demand frontier.
- Peer frontiers are calculated by outlet type and outlet size using high-performing comparable outlets.
- Constraint score combines structural capacity, cooler count, SKU breadth, catchment density, POI demand score, coordinate availability, and plateau behavior.

Final uncap formula

```
raw_potential = lower_bound + uncap_weight * max(peer_frontier - lower_bound, 0)
```

Guardrails

Guardrail	Purpose
Observed lower bound	Avoid predicting below proven outlet capability
Peer 98th percentile soft cap	Prevent unrealistic segment outliers
Outlet-size uplift caps	Limit excessive jumps for small or unknown outlets
Non-negative clipping	Maintain physically valid output

This framework is explainable to business users: high potential requires either demonstrated capability or a defensible route to the peer frontier through structural and catchment evidence.

4. Validation, Outputs, and AI Transparency

Validation item	Result
Platform rows	914
Full output rows	20,000
Missing predictions	0
Mean potential	445.34 L
Median potential	259.77 L
Mean uplift vs observed max	1.36x
Median uplift vs observed max	1.18x
Maximum uplift vs observed max	2.97x

Deliverables

- Platform output: Results/smil_labs_predictions.csv with row_id and Maximum_Monthly_Liters.
- Full business output: Results/smil_labs_predictions_full_20000.csv for all outlets.
- Validation artifacts: Docs/model_validation_summary.md plus top-100 potential and uplift CSVs in data/gold/.

Generative AI transparency

Generative AI was used as an accelerator for structuring documentation, drafting reusable data quality logic, brainstorming latent demand approaches, designing Overpass queries, debugging model and Git issues, and producing validation/report scaffolds. All AI-assisted outputs were reviewed through notebook execution, schema checks, rejected-record summaries, uplift diagnostics, and manual review tables.

Submission caveat

The written challenge brief asks for outlet-level predictions, while the platform validator expects 914 rows. The current platform file uses the 914-row fallback. If the official sample/test template is supplied, rerun the POI enrichment and main pipeline notebooks so the exact official row_id list is used.

Conclusion

The Smil Labs solution provides a reproducible lakehouse pipeline, documented data forensics, external POI integration, an explainable latent-demand model, and validated prediction outputs ready for platform upload and judging discussion.